

# Karim Hajj Ali

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## EDUCATION

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### American University of Beirut

Expected June 2027

*Bachelor of Engineering (B.E.) in Computer Science and Engineering*

*Beirut, Lebanon*

- **Relevant Coursework:** Design & Analysis of Algorithms, Data Structures & Algorithms, Object-oriented & Effective Java Programming, Introduction to Machine Learning, AI in Industry, Database Systems, Linear Algebra, Discrete Structures, Cryptography & Network Security.

## PROJECTS

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### **Ou3a-Joura! — Pothole Detection** *Android sensors — ML — Python, Java*

*Oct 2025 – Dec 2025*

- Developed an ML-based system to detect road potholes from smartphone sensor data.
- Processed accelerometer and gyroscope time-series data collected from Android devices.
- Trained and evaluated models to classify pothole vs non-pothole events.

### **Bankruptcy Risk Prediction** *Python, CatBoost, Docker*

*Sep 2025*

- Built a machine learning system to predict next-year corporate bankruptcy from financial data.
- Engineered time-aware labels to prevent data leakage in year-ahead prediction.
- Trained an imbalance-aware CatBoost model with validation-based threshold optimization.

### **xv6 Operating System Extensions** *System calls, user auth — C, xv6, QEMU*

*Jul 2025 – Aug 2025*

- Implemented custom xv6 system calls and user-level programs to interact with kernel data structures.
- Extended the xv6 boot process by adding a login mechanism using console I/O and file descriptor management.
- Developed and integrated Unix-like user utilities (e.g., **head**) into the xv6 userland build system.

### **Smart Hotel Reservation System** *PostgreSQL — SQL/PLpgSQL*

*Sep 2025 – Dec 2025*

- Designed a relational database for a smart hotel system covering reservations, payments, services, staff, events, and multi-branch operations.
- Built an ERD and mapped it to a normalized schema with advanced relationships (M:N, weak entities, specialization, unary supervision).
- Implemented the SQL schema with constraints, indexes, views, stored procedures, and triggers to enforce business rules and data consistency.

### **AES-128 Encryption Tool** *Python — Tkinter — SQLite — Matplotlib*

*Feb 2025 – Apr 2025*

- Implemented AES-128 encryption/decryption with key expansion and full round transformations.
- Built a Tkinter GUI with user login and per-user encryption history stored in SQLite.
- Added advanced features: round-by-round view, key expansion viewer, and avalanche-effect plots.

## TECHNICAL SKILLS

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**IT:** Python, C/C++, Java, Github, APIs, Flask, scikit-learn, Docker

**Other Skills:** Problem Solving, Team Collaboration, Communication

## ADDITIONAL INFORMATION

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Languages: English (Proficient), French (Intermediate), Arabic (Native)